

Experiment 7

Problem Statement

You are required to design and implement a container-based deployment using Docker to demonstrate infrastructure automation and deployment consistency. Perform the following tasks systematically.

Task 1: Set Up the Working Environment

1. Create a project directory for the experiment.
 2. Navigate into the created directory.
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Task 2: Develop a Simple Web Application

1. Create an index.html file.
 2. Design a basic webpage containing:
 - A heading indicating “Infrastructure Automation using Docker”
 - A short message describing deployment consistency
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Task 3: Automate Infrastructure using Dockerfile

1. Create a Dockerfile in the project directory.
2. Configure the Dockerfile to:
 - Use an official web server image (e.g., Nginx)
 - Copy the HTML file into the appropriate directory inside the container

- Expose the required port
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Task 4: Build the Docker Image

1. Build a Docker image from the Dockerfile.
 2. Assign an appropriate name (tag) to the image.
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Task 5: Deploy the Application Using a Container

1. Run a container from the created image.
 2. Map the container port to a host port.
 3. Access the application through a web browser and verify its output.
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Task 6: Validate Deployment Consistency

1. Stop and remove the running container.
 2. Re-run a new container using the same image.
 3. Observe and record whether the output remains consistent without any changes.
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Task 7: Share the Image using a Public Registry

1. Create an account on Docker Hub.
2. Tag the Docker image appropriately.
3. Push the image to the repository.

Task 8: Questions for Analysis

1. Explain how Docker enables infrastructure automation.
2. Analyze how containerization ensures deployment consistency.
3. Differentiate between a Docker image and a container.
4. Evaluate the advantages of using container-based environments over traditional setups.

Expected Outcome

- The web application should run successfully inside a Docker container.
- The same application should produce identical results across multiple runs.
- The deployment process should be fully automated using the Dockerfile.

Submission Instructions:

1. Students must upload a single PDF file containing the report of the experiment conducted.
2. The report should contain codes/snapshots of outputs/answers corresponding to each task.
3. The PDF name should be in the form "StudentName_SAPID".